THE DIVISOR OF A PRODUCT OF MEROMORPHIC SECTIONS IS THE SUM OF THE DIVISORS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-divisors/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

Editorial note. Statement span 6830-6845 and proof span 6847-6863 come from the halfA pool card operations/pool-cards/stacks-divisors-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The statement's 'In particular' sentence about Weil divisor classes is part of the counted statement span and is preserved.

Allowed standard prerequisites (editorial).

- The definition of the order of vanishing of a meromorphic section at a prime divisor, by comparing the section with a local generator at the generic point of the divisor. Used at proof lines 6851-6859. Supplies the function whose additivity is the content of the proof.
- Meromorphic sections of an invertible module form a module over the sheaf of meromorphic functions, and the product of nonzero meromorphic sections of two invertible modules is a nonzero meromorphic section of the tensor product. Used at statement lines 6834-6839 and at proof lines 6849-6851. Makes the product st a legitimate meromorphic section of the tensor product and lets the local comparison be taken.
- Additivity of the order function on meromorphic functions at a prime divisor. Used at proof lines 6860-6862. Turns the factorisation of the quotient into the sum of the three orders.

Lemma 0.1. *Let X be a locally Noetherian integral scheme. Let $\mathcal{L}$, $\mathcal{N}$ be invertible $\mathcal{O}_X$-modules. Let s, resp. t be a nonzero meromorphic section of $\mathcal{L}$, resp. $\mathcal{N}$. Then st is a nonzero meromorphic section of $\mathcal{L} \otimes \mathcal{N}$, and*

$$div_{\mathcal{L}\otimes\mathcal{N}}(st) = div_{\mathcal{L}}(s) + div_{\mathcal{N}}(t)$$

in $Div(X)$. In particular, the Weil divisor class of $\mathcal{L} \otimes_{\mathcal{O}_X} \mathcal{N}$ is the sum of the Weil divisor classes of $\mathcal{L}$ and $\mathcal{N}$.

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Proof. Let s, resp. t be a nonzero meromorphic section of $\mathcal{L}$, resp. $\mathcal{N}$. Then st is a nonzero meromorphic section of $\mathcal{L} \otimes \mathcal{N}$. Let $Z \subset X$ be a prime divisor. Let $\xi \in Z$ be its generic point. Choose generators $s_\xi \in \mathcal{L}_\xi$, and $t_\xi \in \mathcal{N}_\xi$. Then $s_\xi t_\xi$ is a generator for $(\mathcal{L} \otimes \mathcal{N})_\xi$. So $st/(s_\xi t_\xi) = (s/s_\xi)(t/t_\xi)$. Hence we see that

$$\text{ord}_{Z,\mathcal{L}\otimes\mathcal{N}}(st) = \text{ord}_{Z,\mathcal{L}}(s) + \text{ord}_{Z,\mathcal{N}}(t)$$

by the additivity of the ord_Z function. □